Asmita Sengupta

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Summary

Master's student in Computer Science at RPTU Kaiserslautern-Landau. My research focuses on building robust machine learning systems that work reliably in real-world applications, particularly anomaly detection in time-series data and using large language models to extract and organize information from unstructured text. I'm interested in developing methods that are efficient, interpretable, and trustworthy.

Education

RPTU Kaiserslautern — Kaiserslautern, Germany
M.Sc. in Computer Science, Intelligent Systems (Grade: 1.7) — *Aug 2023 – Jul 2026*

RCC Institute of Information Technology — Kolkata, India
B.Tech. in Computer Science and Engineering (DGPA: 9.26/10, ≈ German grade 1.4) — *Aug 2017 – May 2021*

Experience

Student Research Assistant — Nov 2025 – present
Visual Information Analysis Group, Dept. of CS, RPTU — *Kaiserslautern, Germany*

- Master's thesis (ongoing): Semi-supervised anomaly detection in multivariate time-series data, focusing on robust detection under limited supervision.
- Develop a GUI for data exploration and anomaly detection of multivariate time-series data from a thermodynamic distillation experiment using temperature, pressure, and flow sensors.
- Enable real-time anomaly detection by integrating sensor streams and visualizing key process patterns.
- Evaluate the impact of scoring functions and thresholding strategies on model reliability and generalization across multiple models.

Student Research Assistant - Data Science and its Applications — Aug 2024 – Oct 2025
German Research Center for Artificial Intelligence (DFKI) — *Kaiserslautern, Germany*

- Designed and implemented an end-to-end LLM-based pipeline to transform 13,000+ biomedical drug leaflets into structured knowledge graphs, demonstrating scalable processing of complex unstructured biomedical text with rigorous quality assurance.
- Developed evaluation methodology combining human-in-the-loop validation and LLM-as-a-judge frameworks, achieving **96%** accuracy on downstream entity linking and relation extraction tasks.
- Demonstrated practical utility by applying MEDAKA to excipient-aware drug substitution, a clinically-relevant downstream task enabling pharmaceutical decision support.
- Built a knowledge graph from UK Biobank abstracts to study trends in biomedical research.

Student Research Assistant — Jul 2024 – Jan 2025
Department of Electrical and Information Technology, RPTU — *Kaiserslautern, Germany*

- Developed and evaluated deep learning models for modular reconfigurable batteries, achieving **93%** prediction accuracy across diverse configurations and demonstrating robust generalization in materials science applications.
- Performed systematic data preprocessing, feature engineering, and model evaluation, analyzing performance stability under distribution shift, generalization to unseen battery geometries, and sensitivity to hyperparameter variations.

Security Consultant — Aug 2021 – Aug 2023
PricewaterhouseCoopers India — *Kolkata, India*

- Performed **VAPT** and **IAM audits** for major telecom clients, identifying critical vulnerabilities and supporting overall security improvements.
- Managed **CyberArk** privileged access systems and conducted compliance activities using tools such as **Burp Suite, CyberArk, Nmap, and WireGuard**.

Publications

Sengupta, A., Selby, D. A., Vollmer, S. J., Großmann, G. (2025). *MEDAKA: Construction of Biomedical Knowledge Graphs Using Large Language Models*. arXiv:2509.26128. https://arxiv.org/abs/2509.26128 (Extended version under review at ACM CIKM 2026.)

Sengupta, A., Selby, D. A., Vollmer, S. J. (2026). *Automated Verification of Reproducibility Reporting in AI Research*. Under review at IJCAI-ECAI 2026 CLEAR-AI Workshop.

Skills

AI/ML Concepts: Machine Learning, Deep Learning, Generative AI, NLP, Knowledge Graphs, Anomaly Detection, Model Evaluation, Trustworthy AI, Data Visualization, Visual Analytics

Programming Languages: Python, SQL, Java, C++

AI/ML Frameworks: PyTorch, TensorFlow, Keras, Transformers

Libraries & Tools: NumPy, Pandas, Scikit-learn, Hugging Face, Matplotlib, Seaborn, NLTK, Git, SLURM, D3.js

Projects

Automated Verification of Reproducibility Reporting in AI Research Sep 2025 – Mar 2026
Scientific NLP · Automated Quality Assessment *Python, Hugging Face, GROBID*

• Built a large-scale evaluation pipeline for automated assessment of reproducibility reporting in AI research, combining rule-based NLP, zero-shot LLMs, and hybrid verification methods.

• Systematically designed and evaluated multiple pipeline configurations across different text extraction strategies and LLM configurations, achieving up to 92% accuracy and 0.84 Cohen's kappa on a manually annotated benchmark set, demonstrating robustness of methodology.

• Conducted comprehensive meta-analysis of 3,000+ papers across NeurIPS, ACL, and AAAI to quantify venue-specific and temporal trends in reproducibility reporting standards, contributing to broader discussions on open science in ML.

IMDb Top 1000 Movie Dashboard Apr 2025 – Aug 2025
Interactive Data Visualization · Analytics *Svelte, D3.js, Git, GitHub Pages*

• Built an interactive movie analytics dashboard exploring a century of film data (1920–2020), covering revenue trends, genre distributions, top titles, leading creators, and collaboration networks. Demonstrated end-to-end data visualization system design and implementation.

• Implemented bubble charts, heatmaps, and network graphs using Svelte and D3.js for interactive visual analysis, and built a rule-based recommendation system for suggesting similar movies based on IMDb score, common directors and actors.

Certificates

AWS Certified Cloud Practitioner
AWS Academy ML for Natural Language Processing
AWS Academy Machine Learning Foundations
Generative AI with Large Language Models (Coursera)

Languages

English: Fluent (C1)
German: Intermediate (B1)